

Team Worksheet — Lab 5

CTQ Trees and CTQ Flowdown to Controllable Xs

Team members: _____

Date: _____

1. Our understanding of the problem

2. Our working

Use the data in lab-05-ctq-tree-flowdown.xlsx.

3. What we produced

Deliverable: A CTQ tree with targets and specification limits, plus a flowdown to controllable Xs.

4. What the evidence does NOT tell us

5. Our conclusion

6. Step notes

Step 1. Recap the CTQ tree structure in full: Need -> Drivers -> Measurable Requirements, with each level more specific than the last.

Step 2. Take your top VOC theme and drive it down: what is the need, what drives it, and what specifically must be measured?

Step 3. For each CTQ, write a complete operational definition — what is measured, by whom, with what instrument, at what point, and what counts as a defect.

Step 4. Set the target, the upper and lower specification limits, and the defect definition for each CTQ. A CTQ without limits cannot generate a defect count.

Step 5. Verify each CTQ against the customer: would they agree this number represents what they asked for?

Step 6. Now flow down. Write your business-level Y (e.g. OTIF percentage) at the top of a cascade diagram.

Step 7. Break that Y into sub-Ys — the process-level outputs that roll up into it (pick accuracy, line yield, changeover time, carrier dwell).

Step 8. Break each sub-Y into the Xs that drive it. Continue until you reach Xs a team can directly control or experiment on.

Step 9. Mark on the cascade which Xs you will measure in Day 2, test statistically in Day 3, and experiment on in Day 4.

7. Self-check

Every CTQ has an operational definition, a target and specification limits, and your flowdown reaches at least one layer of Xs that are directly controllable and testable.

Complete? ☐ Yes ☐ Not yet

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